

CO 1: Image Formation

- Camera model
- Camera calibration
- Binocular vision
- 3D reconstruction basics

CO 2: Image Processing

- Image representation
- Edge detection (Sobel, Canny)
- Feature extraction
- Optical flow
- Stereo vision

CO 3: Shape Representation

- Contours
- Region-based methods
- Active contours (Snakes)
- Fourier descriptors

CO 4: Recognition & Learning

- Hough Transform
- PCA
- CNN basics
- Vision Transformers
- SVM
- GMM & EM

CO 5: Applications

- Face detection & Eigenfaces
- Object tracking
- Road sign detection
- Autonomous vehicle perception
- Medical image segmentation
- AR/VR

Important OpenCV programs to

practice Must-do Lab Programs

- Read and display an image
- Convert RGB to Gray
- Image thresholding
- Canny edge detection

- Hough line detection
- Contour detection
- Face detection using Haar Cascade
- Video capture from webcam
- Motion detection
- Simple CNN image classification (optional)

CO I – Introduction to Computer Vision and Image Formation

1. Image Processing

Definition

Image Processing is the process of **improving or modifying an image** using computer algorithms.

Purpose

- Remove noise
- Enhance image quality
- Resize images
- Adjust brightness and contrast
- Prepare images for analysis

Input and Output

Input: Image

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Processing

↓

Output: Improved image

Example

- Removing blur from a photo
- Converting a color image into grayscale
- Increasing image brightness

2. Computer Vision

Definition

Computer Vision is the field of Artificial Intelligence that enables computers to **understand and interpret images and videos**, similar to how humans use their eyes and brain.

Goal

Instead of just improving an image, Computer Vision tries to answer questions like:

- What object is present?
- Where is the object?
- What is happening?
- Who is the person?

Examples

- Face recognition
- Self-driving cars
- Fingerprint recognition
- Medical diagnosis

Computer Vision Pipeline

Image

↓

Preprocessing

↓

Feature Extraction

↓

Recognition

↓

Decision

3. Computer Graphics

Definition

Computer Graphics is the creation of images using computers.

Instead of understanding an image, it **creates** images.

Examples

- Video games
- Animated movies
- 3D modelling
- Virtual worlds

Difference among Image Processing, Computer Vision and Computer Graphics

Image Processing	Computer Vision	Computer Graphics
Improves images	Understands images	Creates images
Input → Image	Input → Image	Input → Model/Data
Output → Better image	Output → Information	Output → Image
Example: Noise removal	Example: Face recognition	Example: Animation

4. Introduction to Computer Vision

Computer Vision allows computers to:

- Detect objects
- Recognize faces
- Track moving objects
- Understand scenes
- Measure distances
- Build 3D models

Basic Steps

Image Capture



Preprocessing



Feature Extraction



Object Detection



Recognition



Decision

5. Levels of Computer Vision

Computer Vision is divided into three levels.

A. Low-Level Vision

Works directly on pixels.

Operations:

- Image enhancement
- Noise removal
- Filtering
- Edge detection

Example

Removing blur from an image.

B. Mid-Level Vision

Extracts useful information.

Operations

- Segmentation
- Feature extraction
- Shape detection
- Motion estimation

Example

Finding the boundary of a car.

C. High-Level Vision

Understands the scene.

Operations

- Face recognition
- Object recognition
- Activity recognition
- Scene understanding

Example

Detecting whether a person is wearing a helmet.

Difference

Low-Level	Mid-Level	High-Level
Improve image	Extract information	AI decisions
	Feature	

6. Applications of Computer Vision

A. Document Image Analysis

Used for:

- OCR (Optical Character Recognition) •
- Handwriting recognition
- Scanning documents

Example

Reading text from a scanned PDF.

B. Biometrics

Identifies people using:

- Face
- Fingerprint
- Iris
- Palmprint

Example

Phone face unlock.

C. Object Recognition

Detects and identifies objects.

Example

Recognizing:

- Car
- Dog
- Chair

D. Object Tracking

Tracks moving objects in videos.

Example

Tracking a football player.

E. Medical Image Analysis

Used in hospitals.

Applications

- Tumor detection
- MRI analysis
- CT scan analysis

F. Content-Based Image Retrieval (CBIR) Searches

images based on visual content instead of text. Example

Upload a flower image and find similar flowers.

G. Video Data Processing

Processes video frame by frame.

Applications

- CCTV surveillance
- Motion detection
- Crowd analysis

H. Multimedia

Uses images and videos in

- Movies
- Games
- Social media
- Video editing

I. Virtual Reality (VR)

Creates a completely virtual environment.

Examples

- VR games
- Flight simulators

J. Augmented Reality (AR)

Adds virtual objects to the real world.

Examples

- Pokémon GO
- Snapchat filters

Difference between VR and AR

Virtual Reality	Augmented Reality
Fully virtual world	Real world + virtual objects
Requires VR headset	Uses mobile camera or AR glasses

7. Camera Models and Image Formation

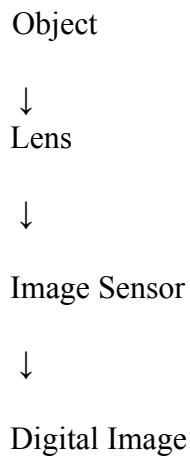
A camera converts a **3D real-world scene into a 2D image**.

Components

- Lens
- Aperture
- Image sensor (CCD/CMOS)

- Image processor

Image Formation Process



8. Camera Model

The **Pinhole Camera Model** is the simplest camera model.

Working

- Light passes through a tiny hole.
- Rays meet at the pinhole.
- An inverted image is formed on the image plane.

Advantages

- Simple
- No lens distortion

Disadvantages

- Dim image
- Not practical

9. Camera Calibration

Definition

Camera calibration determines the camera's internal and external parameters to improve measurement accuracy.

Internal (Intrinsic) Parameters

- Focal length
- Principal point
- Lens distortion

External (Extrinsic) Parameters

- Camera position
- Camera orientation

Applications

- Robot navigation
- Measurement
- 3D reconstruction

10. Binocular Imaging System

Uses **two cameras** like human eyes.

Purpose

Estimate the depth of objects.

Applications

- Robots
- Drones
- Autonomous vehicles

11. Multiple View Geometry

Uses images captured from different viewpoints to determine the **3D structure** of a scene.

Applications

- 3D reconstruction
- Drone mapping
- Photogrammetry

12. Structure Determination

Determines the **shape and position of objects** using one or more images.

Applications:

- 3D scanning
- Robotics
- Industrial inspection

13. Shape from Shading

Recovers the **3D shape** of an object by analyzing brightness and shadows in a single

image. **Example**

A ball appears brighter at the top and darker at the bottom, allowing estimation of its curved shape.

14. Photometric Stereo

Captures **multiple images from the same camera position** while changing the lighting direction.

Purpose

Estimate the surface orientation and fine details of an object.

Difference from Shape from Shading

- **Shape from Shading:** One image, varying brightness.
- **Photometric Stereo:** Multiple images, different lighting.

15. Depth from Defocus

Estimates the **distance (depth)** of an object by analyzing how blurred it appears.

Principle

- Near the focus → Sharp image.
- Far from the focus → Blurred image.

Applications

- Smartphone cameras
- Robotics

- 3D imaging

16. Construction of 3D Models from Images

Creates a **3D model** using one or more 2D images.

Steps

1. Capture images from different viewpoints.
2. Detect common features.
3. Match corresponding features.
4. Estimate camera positions.
5. Reconstruct 3D points.
6. Generate a 3D surface or mesh.

Applications

- Heritage preservation
- Autonomous vehicles
- Medical imaging
- Virtual Reality (VR)
- Augmented Reality (AR)